

# Andreas Eglhofer

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## Education

**Massachusetts Institute of Technology**

Cambridge, MA

B.S. in Physics, Expected 2030

**South Carolina Governor's School for Science and Mathematics**

Hartsville, SC

High School Diploma, 2026

GPA: 4.00/4.00 unweighted, 5.566/5.00 weighted

SAT: 1550, 800 Math, 750 EBRW

Relevant Coursework: Single/Multivariable Calculus, Mechanics, Electricity and Magnetism, Discrete Structures, Number Theory, Modern Physics, Fluids, Thermodynamics, Optics, Data Structures and Algorithms

## Research Experience

**Research Assistant, Max Planck Institute for Software Systems**

Kaiserslautern, Germany

June 2025 – August 2025

- Researched FunSearch-style methods using large language models for combinatorial optimization and search problems.
- Implemented and evaluated approaches on tasks including MaxSAT and word separation.
- Presented technical work in an international research setting under Dr. Rupak Majumdar

## Individual Projects

**Ultimate Tic Tac Toe AI**

Python

- Built an AI for Ultimate Tic Tac Toe using depth-limited minimax with alpha-beta pruning, iterative deepening, and move ordering.
- Designed a heuristic evaluation function combining mini-board control, macro-board line scoring, center and corner weighting, and active-board pressure.
- Implemented state normalization, legal-move generation, terminal-state detection, and time-capped search to improve decision quality under runtime constraints.

**Guard Shift**

andreas-eg.itch.io/guard-shift

- Built a top-down stealth game in Python and Pygame with guard behavior driven by A\* search on a grid-based state space.
- Implemented dynamic path replanning across patrol, investigate, chase, and search states using Manhattan-distance heuristics and walkable-tile successor generation.
- Designed interactive mechanics including noise-triggered guard responses, door-based path blocking, alert escalation, and multi-level gameplay.

**Pressurized Water Reactor Dynamics Simulator**

pwr-simulator.vercel.app

- Built an interactive simulator modeling reactor power output, core temperature, control rod motion, thermal feedback, and SCRAM logic.
- Researched reactor physics independently and translated physical system behavior into a computational model.
- Developed the application in JavaScript and React.

## Technical Skills

**Languages:** Python, C++, Bash, LaTeX

**Libraries:** Pygame, React, NumPy, SciPy, pandas, Matplotlib

**Tools:** Git, Linux, Neovim, Nix, CMake, SSH, KiCad, SolidWorks

**ML:** PyTorch, TensorFlow

## Honors

*President's Cup (2026)* – GSSM's only graduation award, given in place of class rank to the senior who best embodies the school's highest ideals and aspirations.

*Physics Departmental Award (2026)* – Recognized by the Physics Department for exceptional performance and engagement in advanced physics coursework.

*George Eastman Young Leaders Award (2025)* – Selected for academic excellence, leadership, and meaningful contributions to school and community.